SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
Student Name:	**Mohamed Saif A**	**Reg. No.:**	**192521171**
Date:	**27 October 2026**	**Duration:**	60 minutes

CO2 Statement: *Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*

Code of Conduct

I **Mohamed Saif A / 192521171** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Mohamed Saif A

1. SCENARIO STUDY AND ANALYSIS

Project Title: MediQueue – Smart Hospital Appointment & Queue Management System

1.1 Business Domain and Problem Context

Domain: Healthcare administration (outpatient department, OPD, management in a multi-speciality hospital). A typical hospital OPD sees hundreds of patients a day across many departments. Patients arrive early, stand in long registration queues, and then wait for hours with no idea of their turn, while doctors face uneven loads and receptionists rely on paper registers. Appointments by phone are error-prone, double bookings and no-shows are common, and the hospital has no data about waiting times or doctor utilisation. MediQueue replaces this with online booking, a live digital queue and a single patient record.

1.2 Existing Challenges

- **Long waiting times:** patients queue first to register, then again outside the doctor’s room, with no estimate of their turn.
- **Manual booking:** phone and paper bookings lead to double bookings, lost entries and no reminders.
- **High no-show rate:** missed appointments waste doctor time because nobody is reminded or can cancel easily.
- **Uneven doctor workload:** there is no live view of queues, so patients crowd one doctor while another is idle.
- **Scattered records:** previous prescriptions and reports are carried on paper and are often lost.
- **No management data:** the administration cannot measure average wait, footfall or no-show rate.
- **Billing delays:** separate counters for payment and receipts add another queue.

1.3 System Objectives

1. Let patients find a doctor, book, pay and cancel appointments online in under 3 minutes.
2. Reduce the average waiting time in the OPD by at least 30% through a live token queue.
3. Reduce no-shows by at least 25% through reminders and easy rescheduling.
4. Give doctors a patient’s history and give patients their prescriptions digitally.
5. Provide the administration with daily reports on footfall, waiting time and doctor utilisation.
6. Protect patient data with strong security and a complete audit trail.

1.4 Stakeholders

Stakeholder	Type	Needs from the system
Patient / Attendant	Primary user	Book and pay easily, know the waiting time, access history and prescriptions.
Receptionist	Primary user	Register walk-ins, check patients in, manage the queue, issue bills.
Doctor	Primary user	See the day's schedule and queue, view history, record diagnosis, prescribe, order tests.
Hospital Administrator	Primary user	Manage staff, departments and doctors; view reports; ensure backups and audit.
Hospital Management	Sponsor	Better patient satisfaction, higher utilisation, lower overcrowding.
Laboratory / Pharmacy staff	Secondary user	Receive electronic test orders and prescriptions (later release).
Payment and SMS/Email providers	External systems	Process fees and deliver messages.
Regulators / Legal	Oversight	Patient data privacy and record retention.

2. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

2.1 Functional Requirements

36 functional requirements are grouped by module and prioritised with MoSCoW (M = Must, S = Should, C = Could).

ID	Requirement	Pri.
Account and access		
FR-01	A patient shall be able to register with name, date of birth, gender, mobile number and optional email.	M
FR-02	The system shall verify the mobile number with a one-time password (OTP) before activating the account.	M
FR-03	Users shall log in with mobile/email and password or OTP.	M
FR-04	The system shall enforce role-based access for Patient, Receptionist, Doctor and Administrator.	M
FR-05	The administrator shall be able to create, edit and deactivate staff accounts.	M
FR-06	Users shall be able to reset a forgotten password through OTP or email link.	S
Doctors, departments and schedules		
FR-07	The administrator shall manage departments and doctor profiles (speciality, qualification, consultation fee).	M
FR-08	A doctor shall define a weekly schedule, slot length and leave days.	M
FR-09	The system shall show available slots for a doctor on a selected date.	M
Appointments		
FR-10	A patient shall be able to search doctors by department, speciality or name.	M
FR-11	A patient shall be able to book a free slot; the system shall prevent two bookings for the same slot.	M
FR-12	The patient shall be able to pay the consultation fee online (UPI / card) or choose to pay at the counter.	M
FR-13	The system shall generate an appointment ID and a token number and show a confirmation.	M
FR-14	A patient shall be able to reschedule or cancel up to 2 hours before the slot, with the refund rule applied.	M
FR-15	A receptionist shall be able to register a walk-in patient and book an appointment on their behalf.	M
FR-16	The receptionist shall be able to insert an emergency patient at the front of the queue, with a reason recorded.	S
Queue management		
FR-17	The receptionist (or the patient by scanning a QR code) shall be able to check in a patient on arrival.	M
FR-18	The system shall show each patient their token position and estimated waiting time live.	M
FR-19	A doctor shall be able to call the next patient, mark the consultation complete or skip a patient.	M
FR-20	The system shall recalculate waiting times when a consultation runs long or a doctor is delayed.	S
FR-21	The system shall mark a patient as no-show after a 15-minute grace period and release the slot.	S
Consultation and records		
FR-22	A doctor shall be able to view the patient's previous visits, prescriptions and reports.	M
FR-23	A doctor shall be able to record symptoms, diagnosis and notes for each visit.	M
FR-24	A doctor shall be able to issue an electronic prescription with drug, dose and duration.	M

ID	Requirement	Pri.
FR-25	A doctor shall be able to order laboratory tests for a patient.	S
FR-26	A patient shall be able to view and download their history and prescriptions as PDF.	M
Billing		
FR-27	The system shall generate a bill and a PDF receipt for each consultation.	M
FR-28	The system shall record payment status and process refunds for eligible cancellations.	M
Notifications and feedback		
FR-29	The system shall send SMS/email confirmation of booking, rescheduling and cancellation.	M
FR-30	The system shall send reminders 24 hours and 2 hours before the appointment.	M
FR-31	The system shall notify patients when a doctor is delayed or unavailable.	S
FR-32	A patient shall be able to rate the consultation after the visit.	C
Administration and reporting		
FR-33	The system shall show reports of daily footfall, average waiting time, no-show rate and doctor utilisation.	M
FR-34	The administrator shall be able to export reports as CSV or PDF.	S
FR-35	The system shall keep an audit log of logins, record views and changes.	M
FR-36	The administrator shall be able to back up and restore the database.	M

Summary: 28 Must, 7 Should, 1 Could.

2.2 Non-Functional Requirements

Each non-functional requirement carries a measurable target so that it can be tested.

ID	Category	Requirement	Measure / Target
NFR-01	Performance	Booking, search and queue pages shall respond quickly.	95% of requests < 3 s with 1,000 concurrent users
NFR-02	Performance	Live queue updates shall reach patients promptly.	≤ 5 s from doctor action to patient screen
NFR-03	Security	All traffic encrypted; passwords hashed; sensitive fields encrypted at rest.	HTTPS (TLS 1.2+); bcrypt; AES-256 for medical data
NFR-04	Security	Access follows RBAC; sessions expire; every record view is logged.	15 min idle logout; 100% of record views audited
NFR-05	Privacy	Patient data is confidential and used only for treatment and administration.	Compliant with DPDP Act, 2023; no data shared without consent
NFR-06	Reliability	No appointment shall be lost or double-booked, even on failure.	ACID transactions; 0 double bookings in testing
NFR-07	Availability	The service shall be available during OPD hours.	99.9% uptime 7 am – 10 pm; planned maintenance at night
NFR-08	Recoverability	Data shall be restorable after a failure.	Backup every 6 h; recovery point 6 h, recovery time 2 h
NFR-09	Usability	A first-time patient shall book without help; large fonts and simple language for elderly users.	Booking in < 3 min; WCAG 2.1 AA; English and Tamil
NFR-10	Scalability	The system shall add departments and branches without redesign.	Supports 5 branches, 200 doctors and 5,000 appointments/day
NFR-11	Maintainability	Code shall be modular, tested and documented.	≥ 70% unit-test coverage; documented REST API
NFR-12	Interoperability	Data formats shall allow later integration with lab and pharmacy systems.	HL7 FHIR-compatible JSON for patient and prescription data
NFR-13	Compatibility	The app shall work on common devices.	Latest 2 versions of Chrome, Edge, Firefox, Safari; Android and iOS browsers

3. ACTORS AND USE CASES

3.1 Actors

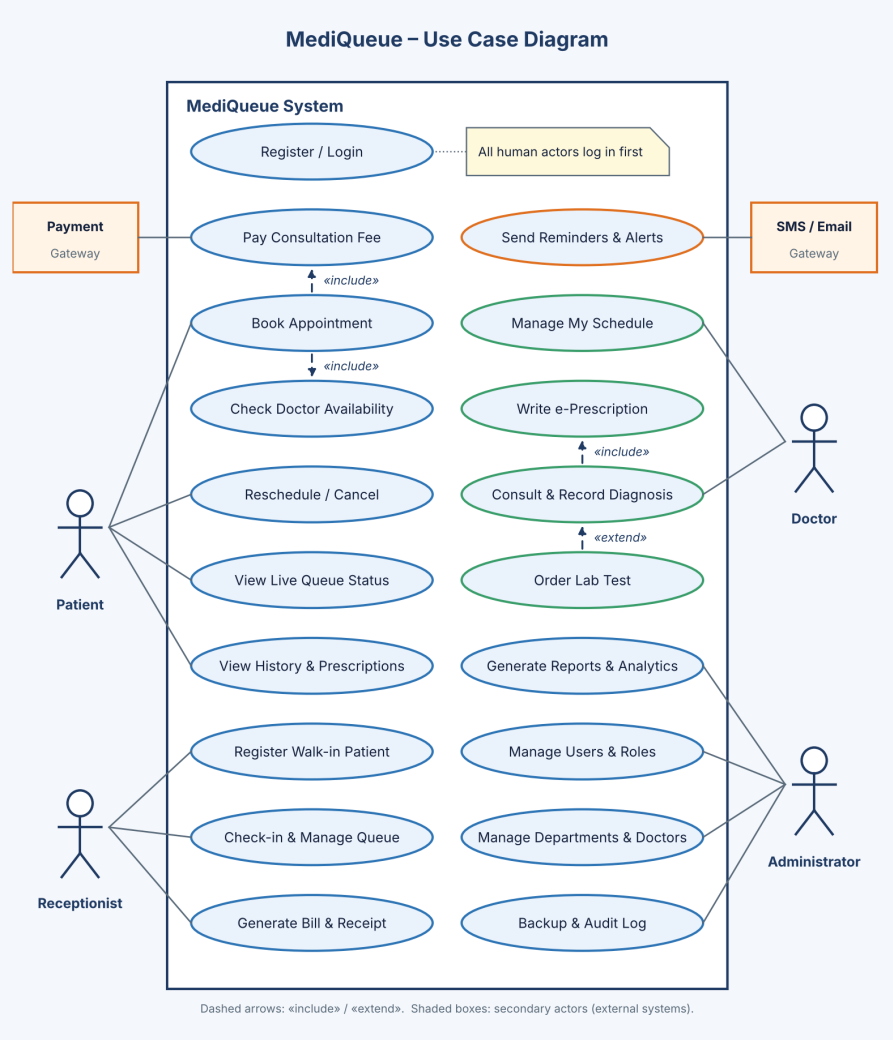
Actor	Type	Description
Patient	Primary	Registers, books and pays for appointments, follows the queue and views records.
Receptionist	Primary	Registers walk-ins, checks patients in, manages the queue and issues bills.
Doctor	Primary	Manages own schedule, consults patients, records diagnosis, prescribes and orders tests.
Administrator	Primary	Manages users, departments and doctors, views reports, handles backup and audit.
Payment Gateway	Secondary	External service that processes online fee payments and refunds.
SMS / Email Gateway	Secondary	External service that delivers OTPs, confirmations and reminders.

3.2 Major Use Cases by Actor

ID	Use Case	Actor(s)	Related FR
UC-01	Register / Login	All human actors	FR-01 – FR-04, FR-06
UC-02	Book Appointment (includes UC-03, UC-04)	Patient	FR-10, FR-11, FR-13
UC-03	Check Doctor Availability	Patient, Receptionist	FR-09
UC-04	Pay Consultation Fee	Patient, Payment Gateway	FR-12, FR-28
UC-05	Reschedule / Cancel Appointment	Patient	FR-14
UC-06	View Live Queue Status	Patient	FR-18, FR-20
UC-07	View History & Prescriptions	Patient	FR-26, FR-32
UC-08	Register Walk-in Patient	Receptionist	FR-15, FR-16
UC-09	Check-in & Manage Queue	Receptionist	FR-17, FR-21
UC-10	Generate Bill & Receipt	Receptionist	FR-27, FR-28
UC-11	Manage My Schedule	Doctor	FR-08
UC-12	Consult & Record Diagnosis (includes UC-13)	Doctor	FR-19, FR-22, FR-23
UC-13	Write e-Prescription	Doctor	FR-24
UC-14	Order Lab Test (extends UC-12)	Doctor	FR-25
UC-15	Send Reminders & Alerts	SMS / Email Gateway (system-triggered)	FR-29 – FR-31
UC-16	Generate Reports & Analytics	Administrator	FR-33, FR-34
UC-17	Manage Users & Roles	Administrator	FR-05
UC-18	Manage Departments & Doctors	Administrator	FR-07
UC-19	Backup & Audit Log	Administrator	FR-35, FR-36

3.3 Use Case Diagram

The diagram follows UML notation: actors outside the system boundary, use cases as ovals, solid lines for associations and dashed arrows for «include» and «extend» relationships.



4. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Introduction

Purpose: This SRS defines what MediQueue must do and how well it must do it, giving the hospital, developers and testers one agreed baseline. **Audience:** hospital management, doctors and reception staff, administrators, developers and testers. **Conventions:** requirements are numbered FR-xx and NFR-xx, use cases UC-xx; priorities follow MoSCoW. **References:** IEEE 830 / ISO/IEC/IEEE 29148; Sommerville, *Software Engineering*; Pressman, *Software Engineering: A Practitioner's Approach*.

4.2 Problem Description

Hospital OPDs rely on phone calls, paper registers and physical queues. Patients wait for hours without knowing their turn, appointments are double-booked or missed, and records are lost. MediQueue is a web platform where patients book and pay online, receive reminders, follow a live token queue, and access their prescriptions, while receptionists, doctors and the administration work from one shared, secure system.

4.3 Scope of the System

In scope	Out of scope (this release)
Patient registration and secure login	
Doctor, department and schedule management	Inpatient (ward) management and bed allocation
Online and walk-in appointment booking	Pharmacy stock and laboratory result processing
Online payment, billing and receipts	Insurance claim processing
Live token queue and wait-time estimates	Telemedicine / video consultation
Consultation notes, e-prescriptions, lab test orders	Native mobile apps (responsive web only)
Patient access to history and prescriptions	Medical imaging (PACS) integration
Reminders, reports, audit log, backup	

4.4 Use Case Descriptions

UC-01 Register / Login

Field	Description
Actor	Patient, Receptionist, Doctor, Administrator; SMS/Email Gateway (secondary)
Precondition	User has a mobile number (patient) or an account created by the administrator (staff).
Main flow	1. Patient chooses Register and enters name, date of birth, gender and mobile number. 2. System sends an OTP by SMS. 3. Patient enters the OTP; account is activated. 4. Later, user logs in with mobile/email and password or OTP. 5. System opens the role's home page.
Alternate flows	A1: Wrong or expired OTP – user may request a new OTP (maximum 3 attempts). A2: Wrong password 5 times – account locked for 15 min.
Postcondition	User has an authenticated session with role-based permissions.

UC-02 Book Appointment

Field	Description
Actor	Patient; Payment Gateway (secondary)
Precondition	Patient is logged in; the doctor has a published schedule.
Main flow	1. Patient searches by department, speciality or doctor name. 2. Selects a doctor and date; system shows free slots (UC-03). 3. Patient selects a slot. 4. System holds the slot for 5 minutes and asks for payment (UC-04). 5. On success, system confirms the booking, generates the appointment ID and token, and sends an SMS/email.
Alternate flows	A1: Slot taken by someone else – system asks the patient to choose another. A2: Payment fails or times out – slot is released. A3: Patient chooses Pay at counter – booking is confirmed as unpaid.
Postcondition	An appointment with a unique token exists and appears in the doctor's schedule.

UC-05 Reschedule / Cancel Appointment

Field	Description
Actor	Patient; Payment Gateway (for refunds)
Precondition	A confirmed appointment exists and the slot is more than 2 hours away.
Main flow	1. Patient opens My Appointments and chooses an appointment. 2. Chooses Reschedule or Cancel. 3. For reschedule, selects a new free slot. 4. System updates or releases the slot, applies the refund rule, and notifies the patient.
Alternate flows	A1: Less than 2 hours remain – system refuses and advises calling the hospital. A2: No free slot – system offers the next available date.
Postcondition	Slot is updated or released; payment status is updated.

UC-09 Check-in & Manage Queue

Field	Description
Actor	Receptionist
Precondition	Patient has a confirmed appointment (or has been registered as a walk-in) for today.
Main flow	1. Patient arrives; receptionist searches by appointment ID or mobile number (or the patient scans the QR code). 2. System marks the patient Checked-in and places them in the doctor's queue by token. 3. System shows the position and estimated wait on the patient's screen and the waiting-hall display.
Alternate flows	A1: Patient is more than 15 minutes late – system marks no-show and offers the next free slot. A2: Emergency – receptionist inserts the patient at the front with a reason (FR-16).
Postcondition	Patient is in the live queue with a token.

UC-12 Consult & Record Diagnosis

Field	Description
Actor	Doctor
Precondition	Doctor is logged in; the patient is checked-in and called.
Main flow	1. Doctor selects Call Next; system displays the patient's token. 2. Doctor reviews previous visits and reports. 3. Records symptoms, diagnosis and notes. 4. Writes an e-prescription (UC-13) and optionally orders tests (UC-14). 5. Marks the consultation complete; the next token is called.
Alternate flows	A1: Patient does not respond to the call – doctor skips; patient moves back 3 places. A2: Consultation runs long – system recalculates waits (FR-20).
Postcondition	The visit record is saved and visible to the patient.

UC-16 Generate Reports & Analytics

Field	Description
Actor	Administrator
Precondition	Administrator is logged in.
Main flow	1. Administrator opens Reports. 2. Chooses the period, department or doctor. 3. System shows footfall, average waiting time, no-show rate and doctor utilisation as charts and tables. 4. Administrator exports the report as CSV or PDF.
Alternate flows	A1: No data for the chosen period – system shows an empty-state message.
Postcondition	Report is displayed or exported; no data is changed.

4.5 Data Requirements

Entity	Key attributes	Relationships
Patient	patient_id (PK), name, dob, gender, mobile (unique), email, address, guardian_id (FK, optional)	Has many appointments and visits
Staff (User)	user_id (PK), name, role, email, password_hash, status	Doctors link to a Doctor profile
Department	dept_id (PK), name, location	Has many doctors
Doctor	doctor_id (PK), user_id (FK), dept_id (FK), speciality, qualification, fee	Has many schedules and appointments
Schedule / Slot	slot_id (PK), doctor_id (FK), date, start_time, end_time, status	A slot has at most one appointment
Appointment	appt_id (PK), patient_id (FK), slot_id (FK), type (online/walk-in), status, created_at	Has one token, one bill
QueueToken	token_id (PK), appt_id (FK), token_no, checkin_time, called_time, status	Belongs to one appointment
Visit	visit_id (PK), appt_id (FK), symptoms, diagnosis, notes	Has many prescription items, lab orders
Prescription / LabOrder	rx_id (PK), visit_id (FK), drug, dose, duration; lab_id (PK), test_name, status	Belong to one visit
Bill / Payment	bill_id (PK), appt_id (FK), amount, mode, status, refund_amount, paid_at	One per appointment
Notification / Feedback	notif_id (PK), patient_id (FK), channel, message, sent_at; rating, comment	Many per patient
AuditLog	log_id (PK), user_id (FK), action, entity, timestamp, ip_address	Written on every login, record view and change

- Mobile number is unique per patient; each slot can hold only one active appointment (enforced by a database constraint).
- Medical records are retained for at least 5 years; audit logs for 3 years. Records are never deleted, only archived.
- Visit and prescription data are encrypted at rest. Estimated storage is about 20 GB per year for 5,000 appointments a day.

4.6 External Interface Requirements

Interface	Requirement
User interface	Responsive web application. Patient screens: search, booking, payment, My Appointments, live queue, history. Staff screens: reception desk, doctor's queue and consultation page, admin console and reports. Large buttons, simple language, English and Tamil, WCAG 2.1 AA.
Hardware	Reception PC with a QR scanner and receipt printer; a wall-mounted screen in the waiting hall showing tokens; patients use any smartphone or computer. Server: Linux VM with 8 vCPU, 16 GB RAM and 500 GB storage.
Software	PostgreSQL database; payment gateway API (UPI / cards, REST/JSON); SMS gateway and SMTP for OTP, reminders and receipts; PDF generator; WebSocket service for the live queue.
Communication	All traffic over HTTPS (TLS 1.2+); REST + JSON API under /api/v1 secured by tokens; WebSocket for live queue updates; messages sent asynchronously through a queue.

4.7 System Features

#	Feature	Description	Priority
1	Secure access	OTP registration, login, role-based permissions	High
2	Doctor and schedule management	Departments, doctor profiles, slots, leave	High
3	Online appointment booking	Search, slot booking, payment, reschedule, cancel	High
4	Live queue management	Check-in, token, wait-time estimate, emergency priority	High
5	Consultation and records	History, diagnosis, e-prescription, lab orders	High
6	Billing and receipts	Bill, receipt, refunds	Medium
7	Notifications and feedback	Confirmations, reminders, delay alerts, ratings	Medium
8	Reports and administration	Analytics, export, audit log, backup	Medium

5. ASSUMPTIONS AND CONSTRAINTS

5.1 Assumptions

- Most patients have a smartphone with internet access; the reception desk helps those who do not.
- Every patient has a mobile number that can receive SMS.
- Doctors publish their schedules in advance and update leave on time.
- The hospital provides the server or cloud account, a payment gateway account and an SMS service.
- Each doctor sees patients in fixed-length slots (for example 10 or 15 minutes).
- The interface is available in English and Tamil in this release.
- Staff receive a short training before go-live.

5.2 Constraints

Type	Constraint
Technical	Web-based only; open-source stack (PostgreSQL, Linux); depends on the payment and SMS providers' availability; must run on low-cost reception PCs.
Operational	Peak load in the morning OPD hours; system must keep working during network breaks at reception (local queue buffer); changes only outside OPD hours.
Economic	Fixed hospital budget; open-source tools; payment and SMS charges per transaction.
Legal	Digital Personal Data Protection Act, 2023 and medical-record confidentiality rules; written patient consent for data use; records retained as required by law.
Time	Prototype in 10 weeks; pilot in one department; full release in one semester (about 16 weeks).
Organisational	Fee and refund rules must follow hospital policy; clinical decisions remain with doctors.

6. REQUIREMENT VALIDATION

6.1 Coverage of Stakeholder Requirements

Stakeholder need	Objective	Covered by
Book, pay and cancel easily (Patient)	1, 3	FR-09 – FR-14; UC-02 – UC-05
Know my turn and waiting time (Patient)	2	FR-17 – FR-21; UC-06, UC-09
Fewer missed appointments (Hospital)	3	FR-14, FR-21, FR-29 – FR-31; UC-05, UC-15
Handle walk-ins and emergencies (Receptionist)	2	FR-15 – FR-17; UC-08, UC-09
See history and prescribe (Doctor)	4	FR-22 – FR-25; UC-12 – UC-14
Own records available to the patient	4	FR-26; UC-07
Management data (Administrator)	5	FR-33, FR-34; UC-16
Security, privacy and audit	6	FR-35, FR-36; NFR-03 – NFR-08; UC-19

6.2 Quality Checks and Findings

Check	Finding	Improvement made / suggested
Ambiguity	“Respond quickly” in NFR-01 and “short wait” in objective 2 were vague.	Replaced by measured targets: < 3 s for 95% of requests and 30% lower average wait than the present baseline.
Ambiguity	FR-16 does not say what counts as an emergency.	Emergency to be declared by the receptionist with a mandatory reason, reviewed by the administrator in the audit log.
Inconsistency	FR-14 allows cancellation until 2 hours before, the same time as the second reminder (FR-30).	Reminder at 2 h includes a Cancel link that stays valid until 2 h before the slot; the later reminder says “final”.
Inconsistency	FR-11 forbids double booking but FR-16 inserts emergency patients.	Emergency patients get a queue position, not a booked slot; slot rules are unchanged.
Redundancy	FR-29, FR-30 and FR-31 all send messages.	Kept as separate requirements but implemented by one notification service (UC-15).
Missing	No rule for children or elderly patients booked by a guardian.	Suggested: add a guardian link (guardian_id) and allow booking on behalf of a dependant.
Missing	No handling for duplicate patient records or insurance details.	Suggested: duplicate check on mobile + date of birth; insurance claim handling kept for a later release.

6.3 Conclusion

Every stakeholder need in Section 1 is traced to at least one functional requirement and one use case, every non-functional requirement has a measurable target, and the assumptions and constraints are stated. The two open items (guardian booking and duplicate-record handling) should be confirmed with hospital management before design begins because they change the data model.

End of Document — This submission addresses all six required aspects of the Scenario-Based Requirement Analysis: scenario study, functional and non-functional requirements, actors and use cases with diagram, the SRS, assumptions and constraints, and requirement validation.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	**/25**